

AMARNATH S. PATEL

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EDUCATION

University of Central Florida

B.S. Physics (Optics & Lasers) and Mathematics, Computer Science minor

Undergraduate Student 4.00 GPA

August 2025 – Expected May 2029

- **Mathematics:** Applied Linear Algebra, Linear Algebra (proof-based, in progress), Complex Analysis, Partial Differential Equations (in progress), Mathematical Methods for Physics, Ordinary Differential Equations, Honors Calculus III, Discrete Structures
- **Physics:** Quantum Information Processing, Modern Physics, Geometric Optics & Lab, Electricity & Magnetism I (in progress), Independent Research

Florida Atlantic University

University coursework via FAU High School, ages 14–18 (111 credit hours) 3.66 GPA

- Deep Learning, Data Structures & Algorithms, Computer Architecture, Computer Logic Design

RESEARCH EXPERIENCE

CELERIS – Independent Research: From-Scratch RCWA Solver & Metalens Design Pipeline

2026

- Built a from-scratch solver for Maxwell's equations in periodic subwavelength structures via rigorous coupled-wave analysis: 1D TE/TM gratings with Li/Liu–Fan inverse-rule factorization, full 2D-vectorial formulation for bi-periodic arrays, and stable Redheffer scattering-matrix recursion.
- Validated every layer of the stack against closed-form physics, an independent method, or energy conservation – RCWA vs. transfer-matrix to $1e-6$, multilayer S-matrix to $1e-15$, 2D → 1D degenerate reduction to $5e-12$, summed diffraction efficiency 1.000000. Independently cross-validated against the gcrewa and Stanford S^4 solvers to $\approx 1e-7$ and reproduces canonical published metalenses (Khorasaninejad et al., *Science* 2016; Chen et al., *Nat. Nanotechnol.* 2018).
- Gradient-based (Adam) inverse design over meta-atom parameter sweeps; Monte-Carlo fabrication-tolerance analysis; Strehl, Zernike wavefront, MTF and chromatic-focal-shift analysis battery.
- Accelerated Rayleigh–Sommerfeld far-field propagation as a CUDA kernel, benchmarked against the project's own optimized 16-core CPU path: $4.8\text{--}5.8\times$ at 92k pillars, agreeing to machine precision. Diagnosed the kernel as memory-bound from the speedup's decline with grid size. C++23/CUDA with Python bindings. *Sole-author manuscript in preparation (Computer Physics Communications).*

Undergraduate Researcher – UCF Astrophotonics Lab, CREOL

August 2025 – Present

Supervisor: Dr. Stephen Eikenberry

- Measuring the wavelength-dependent complex transfer matrix of a photonic lantern by off-axis digital holography: align the interferometric bench and acquire holograms of the multimode output for each input port across the C-band (1525–1575 nm), recovering full amplitude and phase.
- Implemented the [phase-retrieval and mode-decomposition analysis](#): FFT sideband isolation and demodulation, Butterworth low-pass, joint numerical optimization of mode-field diameter, defocus quadratic phase and field position, then decomposition onto the LP basis. Produced the per-port reconstruction-fidelity characterization for 6- and 7-port lanterns at $\approx 98\%$ fidelity against simulated fields.
- Automated the four-instrument acquisition chain (HP 8168E tunable IR laser, Xenics Bobcat 320 InGaAs camera, DiCon GP700 fiber switch, Thorlabs MPC320 polarization control) over GPIB/VISA, RS-232 and GigE Vision, so a complete all-port $\times$ C-band sweep runs unattended with in-loop polarization optimization and saturated-frame rejection – reducing a full dataset from days of manual bench time to a single hands-off run.
- Contribute mount control and pointing automation to [PolyOculus](#), an instrument program building photometric capability from a networked array of 8 telescopes.

Undergraduate Research Assistant – UCF Physics Department

March 2026 – Present

Physics Education Research – Dr. Zhongzhou Chen (funded by NSF Award 2421299; Gates Foundation INV-076932)

- Built [ESTELA](#), generating multi-version isomorphic exams from a structured bank of 615 problems across 29 banks, 13 topic areas and 11 question types, with auto-generated answer keys and export to four formats. Rust (Tauri 2). Materials in use by external adopters; manuscript in preparation.

Undergraduate Researcher – FAU Grant-Funded AI Safety Research Project

January 2024 – March 2025

Supervisor: Tucker Hindle, Florida Atlantic University

- Benchmarked 10 statistical coherence and detection measures (GPT-2 perplexity, RoBERTa, BERT NSP, LSA, NLI, burstiness) on a 255,000-passage corpus; identified GPT-2 perplexity as the strongest discriminator ($3.35\times$ separation, 17.5 vs. 58.5) and showed BERT NSP to be uninformative.
- Built four adversarial evasion pipelines (iterative rewriting, tree-search decoding, list-branching, RL) over 3,125 generated sequences; characterized a consistent quality–evasion tradeoff. Grant-funded with HPC access; presented at the Wilkes Honors College Symposium.

CONFERENCE ABSTRACTS

Sub-diffraction imaging and precision wavefront sensing with photonic lanterns – co-author. Abstract submitted to SPIE Photonics West 2027, OPTO: Photonic Instrumentation Engineering XIV. *Under review.*

TECHNICAL SKILLS

Numerical & Statistical Methods: FFT and signal processing, phase retrieval, modal decomposition, electromagnetic simulation (RCWA), gradient-based optimization, Monte-Carlo tolerance analysis, statistical discriminator benchmarking, GPU/CUDA acceleration

Programming: C/C++ (C++23), CUDA, Rust, Python (NumPy, SciPy), VHDL, Shell (Fish, Bash)

Tools: Linux, Git, CMake, Docker, Qt/PySide6, LaTeX/Typtst; GPIB/VISA, RS-232, GigE Vision instrument control

HONORS & AWARDS

Barry M. Goldwater Scholarship – UCF institutional nominee	2026
Astronaut Scholarship Foundation – UCF institutional nominee	2026
Florida Bright Futures – Florida Academic Scholars (highest tier; 100% tuition)	2025
Lockheed Martin Award, “Highest Level of Engineering Excellence,” AEV Competition	2024
1st Place, Night Hacks Hackathon – UniUtils (course schedule generator & classroom finder)	2023